

Nguyễn Thiện Nhân - 2452879 - HW1

TEKO
SOLUTION

Date: _____

1.

Data Input:

- + Info extracted from envelope
 - + Address
 - + Postcode
 - + Other texts
- Target: Destination / sorting route
- Predicts where the envelope is heading
- Primary metric: Accuracy
- Label noise: incorrectly written or recorded address
- Classification: Predict destination category
- Regression: Predict delivery time

2.

80/20 split $\rightarrow$ 96 for training, 24 for testing. It is simple and provide the estimate of performance on unseen data.

But with only 120 observations, results can depend on what observation enter test set

With 5-fold val on training data + separate held-out test set, CV more stable way to compare models and tune hyperparam. But it still depends on training data, the untouched test set is still needed.

3. $TP = 36$, $FN = 14$, $FP = 20$, $TN = 170$

$\rightarrow \text{Total} = 250$

$$\text{Accuracy} = \frac{TP + TN}{N} = \frac{36 + 170}{250} = 82,4\%$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{36}{36 + 20} = 54,5\%$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{36}{36 + 14} = 72\%$$

$$F1 = \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2(0,545)(0,72)}{0,545 + 0,72} = 62,1\%$$

Accuracy is misleading when ^{Pos} events is rare because it can predict accurately many negative and get high score. False negative needs more attention because they represent real positive events that model fail to classify.

4.

A: 24% train, 26% ^{val} test $\rightarrow$ high, close
 $\rightarrow$ underfit $\rightarrow$ increase model complexity

Date: _____

B: 3% train, 21% val $\rightarrow$ overfit

$\rightarrow$ increase regularization / reduce model complexity

C: 9% train, 11% val $\rightarrow$ low, small gap

$\rightarrow$ good generalization $\rightarrow$ no action needed

$\uparrow$ model complexity $\rightarrow \downarrow$ bias, $\uparrow$ variance

5.-HW1

The project uses the UCI Human Activity Recognition Using Smartphones dataset to classify six activities: walking, walking upstairs, walking downstairs, sitting, standing and laying. The original subject-disjoint train / test split is kept, and validation splits must also separate subjects. Primary metric is macro-F1, with accuracy and confusion matrix as 2nd metrics. A key leakage risk is models using the same split, preprocessing, metrics and random seed (2452819). One limitation is that the dataset covers only specific participants, devices, and activities, so performance may not fully generalize to real-world populations or settings.